

1. Technical Debt Diagnosis

1.1 Identify the hidden-technical-debt categories

(a) Entanglement

- Modifying the logic for one feature altered the predictions for a different feature.
- The feature inputs are tied up within the model; changing one changes how the model uses other features.

(b) Undeclared Consumers

- Marketing reads raw output tables, which creates an unmonitored downstream dependency.
- Things like output rescaling look safe internally but can break externally dependent operations.

(c) Glue-Code Debt / Pipeline Jungles

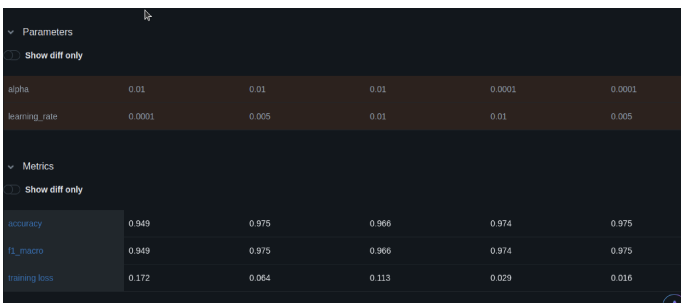
- The training pipeline consists of 14 chained shell scripts with no workflow orchestration or central management.
- Data extraction, preprocessing, and model fitting are linked together manually, making execution difficult to maintain.

1.2 Mitigation for (c)

Tool: MLflow Projects — Consolidate the pipeline into a single documented execution command (`mlflow run .`) runnable from any machine. Define entry points and parameters to replace loose shell scripts and enforce deterministic execution.

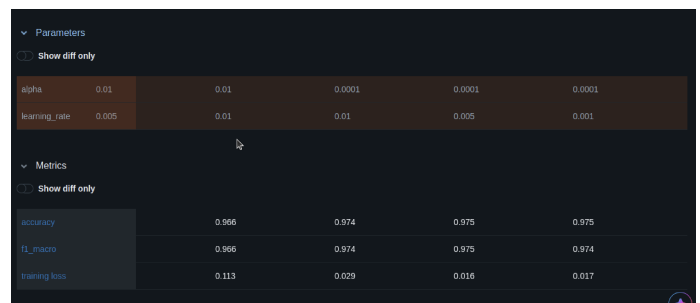
2. Experiment Analysis (MNIST MLP)

- **Best Parameters:** `learning_rate_init = 0.005`, `alpha = 0.0001`.
- **Performance:** Achieved **accuracy** of 0.975 and **f1_macro** of 0.975 with lowest training loss (0.016). This could be due to the decently high learning rate without regularization slowing it down (0.0001).
- **Overfitting:** Minimal evidence of overfitting. Test accuracy increases alongside decreasing training loss (reaching 0.975), proving that for the parameters chosen to be varied, the model generalizes without overfitting too much.
- **Hyperparameter Effect: Learning rate** has a significantly greater impact on model accuracy than **alpha**.
- At `lr = 0.0001`, training loss remains high (0.172) and accuracy drops to 0.949 due to inability to converge.
- Increasing `lr` to 0.005 improves accuracy across all values of **alpha**, dominating overall variance.



Parameters					
☐ Show diff only					
alpha	0.01	0.01	0.01	0.0001	0.0001
learning_rate	0.0001	0.005	0.01	0.01	0.005
Metrics					
☐ Show diff only					
accuracy	0.949	0.975	0.966	0.974	0.975
f1_macro	0.949	0.975	0.966	0.974	0.975
training loss	0.172	0.064	0.113	0.029	0.016

Figure 1: MLflow Run-Comparison Table



Parameters					
☐ Show diff only					
alpha	0.01	0.01	0.0001	0.0001	0.0001
learning_rate	0.005	0.01	0.01	0.005	0.001
Metrics					
☐ Show diff only					
accuracy		0.966	0.974	0.975	0.975
f1_macro		0.966	0.974	0.975	0.974
training loss		0.113	0.029	0.016	0.017

Figure 2: MLFlow Run Table(Extended)